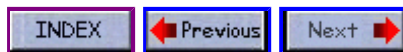


C Programming



GOOD FORM

Perhaps we should say *programming style* or *readability*. The most common complaints we would have about beginning C programmers can be summarized as,

- they have poor layout
- their programs are hard to read

Your programs will be quicker to write and easier to debug if you get into the habit of actually formatting the layout correctly as you write it.

For instance, look at the program below

```
#include<stdio.h>
main()
{
    int sum,loop,kettle,job;
    char Whoknows;

        sum=9;
    loop=7;
    whoKnows='A';
    printf("Whoknows=%c,kettle=%d\n",whoknows,kettle);
}
```

It is our contention that the program is hard to read, and because of this, will be difficult to debug for errors by an inexperienced programmer. It also contains a few deliberate mistakes!

Okay then, lets rewrite the program using good form.

```
#include <stdio.h>

main()
{
    int sum, loop, kettle = 0, job;
    char whoknows;

    sum = 9;
    loop = 7;
    whoknows = 'A';
    printf( "Whoknows = %c, kettle = %d\n", whoknows, kettle );
}
```

We have also corrected the mistakes. The major differences are

- the { and } braces directly line up underneath each other
This allows us to check indent levels and ensure that statements belong to the correct block of code. This becomes vital as programs become more complex
- spaces are inserted for readability

We as humans write sentences using spaces between words. This helps our comprehension of what we read (if you dont believe me, try reading the following sentence.
wishihadadollarforeverytimeimadeamistake. The insertion of spaces will also help us identify mistakes quicker.

- good indentation

Indent levels (tab stops) are clearly used to block statements, here we clearly see and identify functions, and the statements which belong to each { } program body.

- initialization of variables

The first example prints out the value of *kettle*, a variable that has no initial value. This is corrected in the second example.

©Copyright B Brown. 1984-1999. All rights reserved.

